

Interpreting Correlation and Least-Squares Regression

Answer Key

Precalculus / Advanced Statistics

Quick Answers

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|----------|------------------|--------|
| 1. 82 | 5. 100 | 9. 172 |
| 2. 85 | 6. $\frac{1}{2}$ | 10. — |
| 3. 3 | 7. 79 | |
| 4. 0.955 | 8. 4.5 | |
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Curriculum

Level: Precalculus / Advanced Statistics

HSS-IC.A–HSS-IC.B – *problems 9, 10*

HSS-ID.A–HSS-ID.C – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

3. *If they got -3: computed predicted minus observed, which reverses the sign of the residual*
8. *If they got 95.5: reported the percent of variation the model explains instead of the percent it leaves unexplained*
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The data. Practice hours x : 1, 2, 3, 4, 5, 6, 7, 8. Exam scores y : 61, 61, 67, 79, 85, 85, 91, 103.
Least-squares line $\hat{y} = 52 + 6x$, correlation $r = 0.977$.

1. Predict at $x = 5$ hours.

A prediction from a regression line is a substitution, nothing more: put the x -value into the equation the software returned.

$$\hat{y} = 52 + 6(5) = 52 + 30$$

The model predicts 82 points for a student practicing 5 hours per week.

$\hat{y} = 82$

Note the hat: $\hat{y}$ is the model's prediction, not any student's score. The next two problems turn on keeping those two numbers apart.

2. Read the observed score at $x = 5$ hours.

Find the column headed 5 in the table and read the score beneath it. This is a measurement, not a prediction.

85 points

3. Residual at $x = 5$ hours.

A residual is always *observed minus predicted*, in that order, so its sign carries meaning:

$$e = y - \hat{y} = 85 - 82$$

3 points

The residual is positive, so the actual score sits *above* the line: the model **under-predicted** this student by 3 points.

Common wrong turn: predicted minus observed gives -3 and reverses every interpretation — a positive residual would then mean over-prediction.

4. Coefficient of determination from $r = 0.977$.

The coefficient of determination is the square of the correlation:

$$r^2 = (0.977)^2 = 0.954529 \dots$$

Rounded to three decimal places:

$r^2 \approx 0.955$

Interpretation: about 95.5 percent of the variation in exam scores is accounted for by the linear relationship with practice hours. Say “accounted for,” not “caused by” — see problem 10.

Common wrong turn: taking a square root instead of squaring gives 0.988, which is larger than r and therefore impossible: for $|r| \leq 1$, squaring always moves the value toward zero.

5. Predict at $x = 8$ hours, and compare.

$$\hat{y} = 52 + 6(8) = 52 + 48$$

$\hat{y} = 100$

The observed score at 8 hours was 103 points, so the residual is $103 - 100 = 3$ points: this student also beat the model, by the same margin as the 5-hour student. Both points sit above the line, which is visible on the scatter plot.

6. Fraction of points above the line.

Compute each residual $y - \hat{y}$ in order: at $x = 1$ through $x = 8$ the predictions are 58, 64, 70, 76, 82, 88, 94, 100, so the residuals are $+3, -3, -3, +3, +3, -3, -3, +3$. Four are positive.

$$\frac{4}{8} = \boxed{\frac{1}{2}}$$

Why this is expected: the least-squares line is fitted so that the residuals sum to zero. The positive and negative residuals must therefore balance, which puts roughly half the points above the line and half below. A line with, say, seven of eight points above it could not be the least-squares line.

Grading note: accept any explanation resting on “the residuals sum to zero” or “the line passes through the middle of the data.” Exactly half is not guaranteed in general — roughly half is the claim.

7. Mean score, and the line at the mean.

Add the eight scores and divide by 8:

$$\frac{61 + 61 + 67 + 79 + 85 + 85 + 91 + 103}{8} = \frac{632}{8}$$

$$\bar{y} = 79 \text{ points}$$

Now substitute the mean practice time: $\hat{y} = 52 + 6(4.5) = 52 + 27 = 79$ points — exactly the mean score.

The property confirmed: every least-squares line passes through the point of averages $(\bar{x}, \bar{y})$. That is a theorem about the fit, not a coincidence of this data set, and it gives a fast check on any regression equation a calculator hands back.

8. Percent of variation left unexplained.

$r^2 = 0.955$ means 95.5 percent of the variation in scores is accounted for by the model. What is left over is everything else:

$$100 - 100(0.955) = 100 - 95.5$$

$$4.5\%$$

So 4.5 percent of the variation in exam scores is not explained by practice hours — it comes from prior knowledge, test-day conditions, teaching, and ordinary measurement noise.

Common wrong turn: answering 95.5 percent reports the explained share and answers the opposite question.

9. Extrapolate to $x = 20$ hours.

$$\hat{y} = 52 + 6(20) = 52 + 120$$

$$\hat{y} = 172$$

What is wrong with it: the exam is out of 120 points, so a predicted score of 172 is impossible. The arithmetic is correct; the model is being used where it has no evidence. The data only covers 1 to 8 hours, and nothing in it says the relationship stays linear beyond that — in reality returns flatten and the ceiling of 120 points binds.

10. Extrapolation and causation. (*Open response — flagged for manual review.*)

Model answer, part (a). The line was fitted only to practice times between 1 and 8 hours, so it describes the pattern in that window and nothing else. Problem 9 shows the consequence: at 20 hours it returns 172 points on a 120-point exam, an impossible value. A regression model carries no information about behaviour outside the observed range, so predictions there are unsupported no matter how well the line fits inside it.

Model answer, part (b). A correlation of 0.977 says the points lie very close to a straight line; it says nothing about what produced the pattern. This was an observational study — nobody assigned students their practice hours — so a lurking third variable can explain the whole relationship. *Example confounder:* prior mathematical preparation. Students who already understand the material find practice worthwhile and also score well, so preparation drives both variables at once. Other acceptable answers: motivation, hours of sleep, access to a tutor, or the teacher a student happens to have.

Design that would settle it: a randomized controlled experiment. Recruit students, randomly assign each to a required number of practice hours, and compare mean exam scores across the assigned groups. Random assignment balances prior preparation and every other lurking variable across the groups on average, so a difference in outcomes can be attributed to the practice itself.

Grading note: part (a) needs the observed range 1 to 8 hours named and the impossible value used as evidence. Part (b) needs three things: the statement that correlation does not establish causation, one *plausible* confounder that could affect both variables, and a design with *random assignment*. A larger sample or a stronger correlation is not a fix, and an answer proposing one should be corrected — more data from an observational study cannot remove a confounder.